and again the ω_i's will be fixed by momenta that are external to collinear loops. An example where this would not be true is if we had the same collinear direction n in two or more of our building blocks, such as

$$\int d\omega_1\, d\omega_2\; C(\omega_1,\omega_2)\big[\overline{\chi}_{n,\omega_1}\bar{\not{n}}\,\chi_{n,\omega_2}\big]\,. \tag{7.63}$$

For this operator one combination of ω_1 and ω_2 will be fixed by momentum conservation, while the other combination will involve collinear loop momenta. This will lead to anomalous dimension equations of a more complicated form, involving convolutions such as

$$\mu\frac{d}{d\mu}C(\mu,\omega) = \int d\omega'\gamma(\mu,\omega,\omega')C(\mu,\omega')\,. \tag{7.64}$$

Indeed, the operator in Eq. (7.64) is responsible for several classic evolution equations: i) DIS where we have DGLAP evolution for the parton distribution functions $f_{i/p}(\xi)$, ii) hard exclusive processes like $\gamma^*\pi^0 \to \pi^0$ where we have Brodsky-Lepage evolution for the light-cone meson distributions $\phi_\pi(x)$, and iii) the deeply virtual Compton scattering process $\gamma^* p \to \gamma p'$ where the evolution is a combination of both of theses. It is interesting that all of these processes are sensitive to different projections of the evolution of the single operator given in Eq. (7.64). We will carry out an example of an evolution equation with a convolution in the next section, where we consider DIS and the DGLAP equation.

8 Deep Inelastic Scattering

(**ROUGH**) DIS is a rich subject, so for the purpose of these notes we will treat only aspects related to factorization and the renormalization group evolution with SCET. In particular we will demonstrate the factorization of momentum by showing that the forward DIS scattering amplitude can be written as an integral over hard coefficients times parton distribution functions.

8.1 Factorization of Amplitude

The scattering process is depicted in the figure. The hard scale Q of the process is defined by the photon

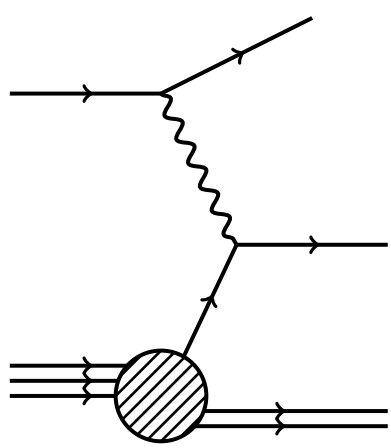

Figure 11: Deep Inelastic Scattering

momentum q^μ

$$q^2 = -Q^2 \tag{8.1}$$